

EE 605 – Error Correcting Codes

Timeslot Mon and Thu 5:30 - 7:00 pm

Web page / Information: Moodle

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Contents:

- Notion of error correcting codes, minimum distance, and rate, error detection / correction capability, basic treatment of algebraic structures like groups / finite fields, linear block codes and their general properties, bounds on code parameters such as rate and minimum distance, popular example constructions such as Hamming, Reed Muller, Reed Solomon codes, concatenated codes, List decoding, Expander Codes.
- *Prerequisites:* Linear Algebra, Probability Theory

Course mechanics:

- Quizzes [30%]
- Midsem [25%]
- Endsem [35%]
- Course project [10%]

References:

- *Error Control Coding*, S. Lin and D. J. Costello, Second Edition, Prentice Hall.
- *Introduction to Coding Theory*, Roth, Cambridge Univ. Press
- *Theory and Practice of Error Control Codes*, Blahut, Addison-Wesley
- *The Theory of Error-Correcting Codes*, MacWilliams and Sloane, North-Holland
- *The Theory of Information and Coding*, McEliece, Cambridge
- *NPTEL Course: Error Correcting Codes*, P. Vijay Kumar,
<https://nptel.ac.in/courses/117/108/117108044/>
- *Video lectures: Error Control Coding*, Andrew Thangaraj,
http://www.ee.iitm.ac.in/~andrew/videlectures/EE512_2010/index.html